# BRENDAN LOU S. MILLARES

*Speech & Audio AI · Engineering & Analytics · Student Leadership ·Creative Leadership*

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## EDUCATION

**De La Salle University – Manila** 2023 – Present
*Gokongwei College of Engineering · Undergraduate*
*Bachelor of Science in Computer Engineering ·Minor in Financial Risk and Analytics*
*DOST-SEI Merit Scholarship*
*St. La Salle Financial Scholarship*
*Dean's Lister — 5 terms (A.Y. 2023–2024 to A.Y. 2025–2026)*

**Valenzuela City School of Mathematics and Science** 2021 – 2023
*Senior High School · Academic Track, STEM Strand*
*Gawad Pagkilala sa Natatanging Kontribusyon sa Larang ng Pamumuno*
*Special Advanced Curriculum with emphasis in the fields of mathematics and science*

**Valenzuela City School of Mathematics and Science** 2017 – 2021
*Junior High School*
*Special Advanced Curriculum with emphasis in the fields of mathematics and science*

**Montessori Academy of Valenzuela** 2007 – 2017
*Kinder to Elementary*

## PROJECTS & RESEARCH

**ECHO: Edge-based Communication for Hierarchical Offline Intent Recognition in Inclusive Atypical Speech Interfaces Using Few-Shot Learning with E-Branchformer and Prototypical Networks**
*— Ongoing Undergraduate Thesis* *Raspberry Pi 5, Few-Shot Learning*

- Designing a portable, offline AAC system running on Raspberry Pi 5 with a ReSpeaker XVF3800 mic array, using few-shot learning (E-Branchformer + Prototypical Networks) to adapt to individual dysarthric speakers

**HimigTransform — A Noise-Robust Audio Fingerprinting System for Identifying American and OPM Songs from Short Audio Segments**
*— Contributing Author · Group Project* *MATLAB, DSP, App Designer*

- Extends Wang's landmark-based constellation fingerprinting (the Shazam-style algorithm) with two enhancements: SNR-adaptive peak picking combined with spectral-subtraction denoising to preserve genuine spectral landmarks under noise, and a short-query hashing mode with a length-normalized decision threshold for 3–5 second clips
- Designed for evaluation on a 100-song reference database (50 American, 50 OPM) across multiple SNR levels and query lengths
- Targeting a ≥10-point accuracy gain over baseline at 0 dB SNR, ≥85% accuracy on 5-second clean queries, and sub-second matching time
- Packaged in a MATLAB App Designer GUI for live demonstration — the first documented benchmark of Shazam-style fingerprinting on OPM repertoire

**Development of a Fine-Tuned Hybrid Classical–YOLOv8 System for Wiring Defect Detection on Single-Layer PCBs**
*— Group Research Paper · IEEE-Format Conference Paper* *Python, OpenCV, YOLOv8s*

- Enhances a hybrid classical + YOLOv8 PCB inspection pipeline with a golden-reference diff stage: ORB feature matching and RANSAC homography align a test board image to a known-good reference
- A pixel-level diff flags added copper (shorts) and missing copper (opens) that a purely self-referential structural stage cannot reliably catch
- Written up as an IEEE-format conference paper

### A YOLOv8-Based Computer Vision System for Automated Crack Detection and Dimension Measurement

*— Contributing Author · Group Project* *YOLOv8, MobileNetV3, ResNet50, EfficientNet*

- Group computer vision pipeline pairing YOLOv8 with MobileNetV3, ResNet50, and EfficientNet classifiers for concrete crack detection and dimension measurement
- Developed on Windows / RTX 4050

### AI-Powered Occupational Safety Risk Prediction System

*— Contributing Author · Group Project* *YOLOv8, MobileNetV3, ResNet50, EfficientNet, AI, PPE Monitoring*

- YOLOv8n/YOLOv8s detection and four classification backbones (MobileNetV3-Small, ResNet50, EfficientNet-B0, baseline CNN) for PPE compliance, with MobileNetV3-Small topping both tasks (98.68% helmet / 93.64% glove accuracy) at a fraction of ResNet50's size.
- Ran a SHAP-based leakage audit that caught an overfitted risk model (99.27% acc. from a leaked feature) and corrected it to an XGBoost classifier at 0.9817 ROC-AUC.
- Grounded the whole pipeline in Philippine labor data (RA 11058, PSA ISLE), surfacing that gloves — the PPE class tied to the most local injuries — were also the hardest to detect (0.17–0.27 mAP vs. 0.54–0.74 for helmet/shoe)

### BEAT-BIT — Modular Heart Monitoring System

*— Contributing Author · Group Project* *TinyML, ESP32, Edge Impulse*

- Co-authored wearable ML system (MAX30102 + ESP32-C3) classifying activity states with an on-device TinyML model (Edge Impulse)
- Achieved 89.3% validation accuracy at 68 ms latency

### Design and Simulation of a 3-Band Real-Time Audio Equalizer Using IIR Butterworth Filters

*— Contributing Author · Group Project* *MATLAB, Signal Processing Toolbox, Audio Toolbox*

- Built a MATLAB app with a live 3-band (Bass/Mid/Treble) IIR Butterworth equalizer supporting real-time playback and live microphone recording
- Adjustable gain sliders (−12 to +12 dB) with per-band mute, master volume boost up to 200%, and seek/transport controls
- Live frequency-response and energy-split visualizations

### Building an Integrated Financial Database Using SQL in R (FINSTAD Case 1)

*— Contributing Author · Group Project* *R, dplyr, SQL-Style Joins, FRED / Yahoo Finance Data*

- Built an integrated financial database in R using dplyr as a SQL-equivalent join/query framework, merging eight datasets (PLTR, SPY, ACEN, SOL-USD, plus four FRED macro series) onto a single trading-day timeline via a left-join anchored on SPY's calendar
- Ran ten SQL-style financial investigations covering calendar-year returns, beta/correlation, volume-volatility relationships, seasonality, weekend-vs-weekday effects, regime decomposition, Sharpe ratios, maximum drawdown, and performance across inflation/VIX/rate regimes
- Translated the findings into a recommended $1,000,000 allocation (30% PLTR, 20% SOL, 15% ACEN, 35% cash) with full portfolio-level backtesting and a stress-test validation

### The Scholar Hive

*— Contributing Author · Group Project* *PHP, MySQL*

- Admin/user scholar-profile management system with search & filtering by scholarship type/status, an automated + admin-triggered email system, profile-photo/avatar handling, and a file-submission feature (PDF requirement uploads)
- Role: admin dashboard, scholars module, email-blast; file-submission feature and email-based prompts; formatting/syncing between admin and user views; debugging; UI design; provided LSS domain details

### DLSU CSO LEAP 2026 Website

*— Front-End Development* *React, TypeScript, Framer Motion*

- Focused on front-end implementation: built UI components in React/TypeScript with Framer Motion animation
- Followed a dark green/gold palette and Tropikal/Playfair Display typography
- Handled iterative UI/UX fixes based on design review

#### OTHER PROJECTS

- **G-Scan** — Python/data tool converting GCash receipts into Excel format
- **Plantita-Fi** — NodeMCU/Blynk IoT system for smart hydroponics; environmental monitoring and control for urban planting
- **BedChem** — Discrete Math project applying Fractional Knapsack and Dijkstra's Algorithm to balance clinical health evaluation and resource distribution efficiency
- **ELM Prestige** — Web design/dev for The Signature Beauty Network's official site

## EVENTS & OPERATIONS

**LSSUMMIT 2026** May 2026 – August 2026

*— Core Leader for Events*

- Built the Day 2 master script (hosts + segment hosts) and handled the execution aspect of entire event.
- Designed the poster judging rotation system (SIM/CHP judges, 9 rounds, 19 tables)
- Coordinated CREAxFIN printables (bingo cards, certificates, tarpapel worksheets, tent signage)
- Restructured the three-part Pre-Summit playbook
- Exploring an Alpha/Bravo/Charlie OpCom cell structure for cross-functional management

**Transcend: "ETHEREAL" — The Scholars' 3rd General Assembly** July 2026

*— Vice President for Activities / Project Head*

- Culminating and celebratory General Assembly bringing together scholars, officers, partners, and communities to highlight the organization's collective journey and growth
- Used interactive activities, committee-led engagement spaces, multimedia presentations, and community-centered segments to promote unity, visibility, and meaningful scholar interaction

**LEAP 2026: PaLSSiklaban** June 2026

*— Vice President for Activities / Project Head*

- A fun, friendly intra-LSS competition where scholars engage in classic Filipino games testing physical strength and mental agility, grounding academic excellence in cultural identity, community engagement, and holistic well-being
- Structured as a committee battle inviting scholars to join specific committees
- **Contribution:** built the event scoring spreadsheet (Amazing Race + Tug of War tabs, formula auditing)

**Lemniscate: "EXTRAVAGANZA" — The Scholars' 2nd General Assembly** April 2026

*— Vice President for Activities / Project Head*

- Culminating event of Scholars' Week, bringing together scholars to reflect, reconnect, and move forward as one community
- Anchored on the lemniscate (∞) symbol of continuity and growth; retracked the week's accomplishments and realigned scholars with LSS's vision and direction

**RippLSS 2.0: Scholars' Forum on Corruption and Development** March 2026

*— Vice President for Activities / Project Head*

- A three-hour interdisciplinary forum engaging Lasallian scholars in critical discussions on corruption, governance, and development issues
- Culminated in scholarly outputs such as policy notes and reflection briefs
- Also served as a platform to introduce Re-calLSS

**Alpha: ENCORE — The Scholars' General Assembly** November 2025

*— Vice President for Activities / Project Head*

- Aimed to build a stronger sense of community and belonging among LSS scholars by creating a welcoming environment for connection and relationship-building
- Informed scholars about the society's key initiatives, upcoming events, and ways to get involved
- Beyond a gathering, established a foundation and declaration of identity and potential partnerships for Lasallian Scholars

**RippLSS: Youth in Action for Civic Duties and Social Responsibilities** August 2025

*— Post-Election LSS Seminar / Project Head / Vice President for Activities*

- Designed to re-engage students in civic and social involvement beyond the electoral period
- Addressed the uncertainty and hesitation students feel about their role in social change once election-season energy fades

**Life LSSons** Year-Long Series

*— Podcast / Videocast Series / Vice President for Activities / Project Head*

- Scholars and alumni share stories, insights, and conversations reflecting both the serious and lighter sides of being a Lasallian scholar
- Goes beyond "a day in the life" narratives to cover national issues, personal growth, organizational life, and casual banter
- Integrates **Re-calLSS** as a subseries extending podcast discussions into timely reflections on current events

**LSSUMMIT** Oct 2024 – Sept 2025

*— Lead Associate for Events*

- Led coordination of **program flow, hosting, and on-ground execution** for the Lasallian Scholars Summit

**SYNTAX Math Challenge** Sept 2024 – June 2025

*— Vice President for Activities*

- Flagship engineering event overseen as part of that role

- During this term, also involved with:
  - **VoyageX 123** — A DLSU SYNTAX outreach event inviting incoming freshmen of the Gokongwei College of Engineering (GCOE) to join the term's flagship gathering, framed around a nautical “set sail together” theme meant to build camaraderie, solidarity, and a shared sense of what it means to be a Lasallian Engineer
  - **ONE-2-4 ALL: Trailblazing Together** (October 23, 2024) — A guest-speaker forum held at Yuchengco 301 for students still uncertain about their path as future Lasallian Engineers; two guest speakers shared insights and personal experiences to guide attendees through their engineering journey, paired with interactive activities
  - **BoardENG** (November 16, 2024) — A Zoom seminar sharing exam strategies from DLSU's top engineering board exam achievers, covering study techniques and stress-management methods to help students approach their board exams with confidence

## LEADERSHIP

**DLSU Green Media Group (GMG)** Sept 2025 – Present

*— Rookie & Member for Social Media Management*

- Assist in creating and scheduling social media content aligned with GMG’s branding and campaigns
- Support engagement efforts by responding to comments, monitoring interactions, and analyzing basic post performance
- Collaborate with senior creatives and fellow rookies to brainstorm concepts for upcoming projects and digital initiatives

**DLSU Lasallian Scholars Society (LSS)** Sept 2025 – Sept 2026

*— Vice President for Activities*

- Plan and facilitate activities and events that enrich scholars’ academic and personal growth
- Maintain communication channels to keep scholars informed and engaged through events
- Coordinate attendance and participation to maximize scholar involvement

**DLSU USG Office of the Vice President for Internal Affairs (OVPIA)** Dec 2025 – April 2026

*— Team Leader for Media Relations (Logistics), Animo Pride 2026*

- Managed media partnerships for Animo Pride 2026, coordinating with internal publications (The LaSallian, Arch Network) and external press (Manila Times) for event coverage

**DLSU USG Office of the Vice President for Internal Affairs (OVPIA)** Jan 2025 – Sept 2025

*— Associate for Community Engagement*

- Supported outreach and service projects within the DLSU community (e.g. DLSU Hall of Fame)

**DLSU Engineering College Government (ECG)** Jan 2025 – Sept 2025

*— Executive Officer for Sports Committee*

- Strategize and execute sports programs that enhance wellness, teamwork, and school spirit
- Coordinate event logistics, facility bookings, and student-athlete engagement
- Serve as liaison between student representatives and university sports administration

**DLSU Data Science Society (DSS)** Oct 2024 – Sept 2025

*— Marketing Executive*

- Strategize marketing content and craft captions for social media campaigns
- Engage audiences through focused and consistent brand messaging

**DLSU Lasallian Scholars Society (LSS)** Oct 2024 – Sept 2025

*— Junior Officer for Scholar Services*

- Assist in planning and executing scholar-focused events and academic support sessions
- Maintain communications channels, coordinate attendance, and manage feedback
- Ensure seamless logistics to support scholar welfare initiatives

**DLSU 78th ENG Batch Government** Jan 2024 – Dec 2024

*— Executive Officer for Publicity & Promotions*

- Developed promotional strategies and materials for batch events and updates

**DLSU Society for Young Engineers Towards Achieving Excellence (SYNTAX)** Sep 2023 – Aug 2024

*— Activities Officer*

- Assisted in planning and running SYNTAX academic, social, and technical events
- Helped organize logistics, venue setup, and attendee registration
- Evaluated event outcomes to refine future activities

## RESEARCH EXPERIENCE

### Research Student (Quantitative)
2017 – 2021

*— Valenzuela City School of Mathematics and Science · Grade VII–XII*

- *Dalandan Peels as an Alternative Ingredient for Hand Sanitizer* — assigned as **Leader** (Grade VII, 2017–2018)
- *The Effectiveness of Amaryllis Belladona (Amaryllis) Pigment as an Alternative Component in Highlighter Pen Ink* — assigned as **Leader** (Grade VIII, 2018–2019)
- *Dalandan (Citrus Aurantium L.) Seeds as a Flocculant in Areas Near Polo River in Valenzuela City* (Grade IX, 2019–2020)
- *Design and Development of Face Shields from Wastepaper Acetate-based Bioplastic Film Sheet and Bamboo Frame* — assigned as **Leader** (Grade X, 2020–2021)
- *Development and Validation of Vehicle Exhaust Pipe Filter using Activated Alumina (Aluminum Oxide) from Waste Cans* (Grade XII, 2021–2022)

### Research Student (Qualitative)
2021 – 2023

*— Valenzuela City School of Mathematics and Science · Grade XI–XII*

- *To L-earn Lifestyle: Valenzuela STEM Students' Experiences of Earn-Life Balance through Online Class during the COVID-19 Pandemic* (Grade XI, 2021–2022)
- *Case 1+4=3?: A Case Study in Developing an Approach through Traditional Counting and Gaming System on Children Diagnosed with Dyscalculia* — Inquiries, Investigation, and Immersion, June 2023; group research project
  - Explored a program-based therapy approach for children with dyscalculia using Traditional Counting Systems and Traditional Gaming Systems (dominoes, sungka), grounded in the Orton-Gillingham multisensory method and the Speedline-Dots Culia approach

## WORK IMMERSION

### Work Immersion (Work-Based Learning) — IT Support
May 17 – 30, 2023

*— Our Lady of Fatima University (OLFU), Valenzuela · IT Department*

- Completed an 80-hour Work Immersion from May 17 to May 30, 2023
- Certificate of Completion signed May 31, 2023 by Biehn Angelo M. Lanuza, IT Staff, IT Department

## CERTIFICATIONS & TRAININGS

### Python Essentials 1
September 30, 2026

*— Cisco Networking Academy*

### Module 1 (2023) – Introduction to Research Ethics
September 10, 2026

*— TRREE (Training and Resources in Research Ethics Evaluation)*

- TRREE training programme in research ethics evaluation; approved by SIWF / ISFM for Continuing Education Programs (2 credits) · CID: 5VIXyqshSY

### Programming Constructs
May 16, 2026

*— MathWorks Training Services · Self-paced training course, 100% completed*

### MATLAB Onramp
May 12, 2026

*— MathWorks Training Services · Self-paced training course, 100% completed*

### Basic Social Media Management Training
August 4, 2023

*— Skill Bridge · Certificate of Attendance · 3-day training via Skype*

- Certificate signed by Retchelie Casera, CEO & Founder

## MULTIMEDIA & CREATIVE

- **ANONAS** — Publicity materials/graphic design for a T-shirt merchandise campaign
- **Muni-Muni** — Short film featuring the literary works of Jerry Gracio (video editing)
- **Fast & Festive** — “Fast, Festive, and Fire” interview special for DLSU Green Media's Animo Christmas production

## TOOLS & SOFTWARE

- **Design & Creative Tools:** AutoCAD, Adobe Flash, Capcut, Davinci Resolve, Canva
- **Engineering / Simulation:** Multisim, Logisim, MATLAB
- **Document Generation:** DOCX/PPTX manipulation (python-pptx, docx Node.js library)
- **Data & Analysis:** R Markdown, SQL, Python-based CV pipelines
- **Languages & Frameworks (from projects):** Python, R, PHP, MySQL, React/TypeScript, OpenCV, YOLOv8

- **Hardware & Embedded:** Raspberry Pi 5, ESP32, NodeMCU, Blynk, Edge Impulse (TinyML)

## ACADEMIC AWARDS & DISTINCTIONS

- **With High Honors** – Rank 10 (Grade XI) *2022*
- **Salutatorian** (Grade X) *2021*
- **With Honors** (Grade IX) *2020*
- **1st Place**, Digital Photography Contest *2022*
- ASEAN–Hiroshima Eco-School in VCSMS *2022*
- Global Center for Good Governance in Tobacco Control Participation *2022*
- Intramurals — **2nd Place**, Doubles Badminton (Grade 12); **2nd Place**, Doubles Badminton (Grade 6)
- Chemical Engineering Society's Absolute Zero Quiz Bee — **Champion** *2024*

## AFFILIATIONS

- **Robotics Guild** — Business Manager (2021–2023); Member (2020–2021) *2020 – 2023*
- **PSYSC Regional NCR Council** — Member *2022 – 2023*
- **YES-O Club** — Member *2017 – 2020*
- **Classroom Officers** — Vice President (2023); Treasurer (2015–2016, 2018–2019) *2015 – 2023*